

CURRICULUM VITAE

Dr Mueen Uddin (Senior Member, IEEE)

Associate Professor of Data and Cybersecurity

College of Computing & Information Technology
University Of Doha For Science & Technology, Doha,
Qatar



Top 2% Research Scientists (Stanford and Elsevier reports 2022, 2023, 2024, and 2025).

Listed among the Top Computer Science Scientists in Qatar, ranked 26th nationally by Research.com

Scopus Author ID: **42062605100**

Google Scholar: <https://scholar.google.com/citations?user=ULNFqZIAAAAJ&hl=en>

ORCID ID: <http://orcid.org/0000-0003-1919-3407>

Scopus: <https://www.scopus.com/authid/detail.uri?authorId=42062605100>

LinkedIn: <https://www.linkedin.com/in/mueen-uddin-bbba4b16/>

Github: <https://mueenuddin-9516.github.io/portfolio/>

Phone:

Mobile: (+974) 66025403

Email:

Mueen.malik@ieee.org

Mueen.uddin@udst.edu.qa

PROFESSIONAL CAREER OUTLINE

Currently, I am Professor of Data and Cybersecurity at the College of Computing and Information Technology, University of Doha for Science & Technology, Doha, Qatar. My primary motivations for employment are teaching, research, and community service. I have maintained balance by demonstrating strong competency in Research (60%), Teaching (30%), and Community Service (10%). My curriculum vitae thoroughly details my teaching and research activities, achievements, honors, and summarizes my prominent academic career accomplishments.

I completed my PhD in Information Systems at Universiti Teknologi Malaysia (UTM), one of Malaysia's leading technology universities, in 2013. My doctoral thesis was titled "Energy Efficient and Low Carbon Enabler Green IT Framework for Data Centers in Pakistan," where I proposed an implementation framework to evaluate and deploy energy-efficient cloud data centers. The findings were published in a Tier-1 journal, *Renewable & Sustainable Energy Reviews* (IF-14.9), showcasing my research as the most comprehensive and rigorously tested measurement framework for cloud data centers.

Throughout my academic career, I have gained international recognition through innovative research, strong leadership, and hands-on teaching in Cybersecurity, Digital Forensics, Business Information Systems, Blockchain and Distributed Ledger Technology, Information & Network Security, and Cloud Computing. This is shown by (i) consistent excellence in teaching and research, with scores above **90%** in faculty performance evaluations over the past eight years at various universities in Malaysia, UAE, KSA, Pakistan, and Brunei, (ii) **over 200** journal and conference publications, (iii) collaboration with several active research groups in Cybersecurity and Blockchain across multiple universities, (iv) serving as a cybersecurity expert and consultant for various universities and related consultancies, (v) acting as Principal Investigator (PI) for multiple research grants in partnership with leading academic and industry colleagues, (vi) being an invited speaker by academic and professional audiences, and (vii) engaging in extensive editorial, review, and academic service activities, as detailed below.

- Throughout my professional research and teaching career, I have published over **220** papers in reputable Scopus and SCI-indexed journals and conference outlets. The number of publications and grants highlights my research excellence, while the awards and accolades reflect the quality of my work. Notably, more than **180** of my publications are in Q-ranked journals with a combined Impact Factor exceeding **400**. Many of my papers, presented at international conferences, have received best paper awards for their significant contributions to the CS and IS communities, particularly in Cloud-based Energy-Efficient Computing. Demonstrating the rigor and impact of my research, **Google Scholar** reports over **9130 citations**, an h-index of **51**, and an i10 index of **112**. Additionally, Scopus (Elsevier) records approximately **6260 citations** and an h-index of **41**.
- As a key indicator of my research profile and impact, I have been listed among the top **2% of research scientists** globally by Stanford University and Elsevier for 2022, 2023, 2024, and 2025. In addition, I am included in the **Research.com** Ranking of Computer Science Scientists in Qatar, with a national rank of 26th. These recognitions highlight the quality, visibility, and international impact of my research contributions within the global computer science and cybersecurity research community.
- I serve as Principal Investigator for the project "**FSC06-0203-260018**", titled "**Next-Generation Protein-Rich Azolla Systems: IoT-Controlled Cultivation, AI Yield Prediction, Post-Harvest Quality Enrichment, and Blockchain-Enabled Sustainability for Qatar Food Security Initiatives**". The project aims to develop a sustainable, locally grown, protein-rich Azolla feed system to reduce Qatar's heavy reliance on imported livestock feed and directly support the food security goals of Qatar National Vision 2030. I lead Work Package 2 (WP2), where we are building an intelligent monitoring system for Azolla ponds using real-time sensors and cameras, and I am developing AI models that analyze plant images to estimate biomass yield and detect early signs of pests or disease before they become visible. I also co-lead Work Package 4 (WP4), where we integrate blockchain technology to create a tamper-proof, transparent record of water usage, harvest data, and sustainability metrics, ensuring that farmers, government agencies, and consumers can trust the environmental claims. Together, AI-powered

monitoring and blockchain-verified transparency guarantee that the Azolla we produce is not only abundant and nutritious but also sustainably and responsibly grown.

- I serve as Principal Investigator for the project "**FSC06-0317-260116**" titled "**SHIELD-Q: AI-Powered Pathogen Surveillance and National Disease Database for Smart Greenhouse Management in Qatar**", a three-year project that protects Qatar's greenhouse vegetable production (targeting 70% self-sufficiency by the end-of 2026) from plant diseases – the sector's biggest biosecurity threat. We built the first Qatar National Plant Disease Database using IoT sensors and a dual-modality imaging system (an autonomous ground robot and a wearable camera) that creates spatially referenced, time-series records of individual plant growth. I help develop an AI detection engine that spots pre-symptomatic disease from plant images, then fuse environmental data with disease probabilities into a unified risk framework. The system generates early alerts (7–14 days before visible symptoms) with controlled false alarms. Finally, we deliver a digital platform with a farm dashboard and a farmer mobile app. Together, SHIELD Q enables data-driven, early disease intervention to secure Qatar's vegetable self-sufficiency targets.
- In 2025, I am involved in a research project titled "**Q-POPs: Systematic Monitoring of Qatar's Persistent Organic Pollutants to Comply with the Stockholm Convention and QNV-2030's Environmental Stewardship Targets**," which aims to assess and manage Persistent Organic Pollutants (POPs) in Qatar to meet both international and national environmental goals. A key component of the project is the development of a private Blockchain platform using Hyperledger technology to establish the Qatar Comprehensive Database for Environmental Contaminants (Q-CDEC). This Blockchain-based system will securely store and process data on POPs collected from environmental samples, such as soil, water, sediments, and vegetation, using advanced techniques like Gas Chromatography-Mass Spectrometry (GC/MS). The platform will guarantee transparency, immutability, and tamper-proof data storage, supporting long-term environmental management, research, and policymaking, helping Qatar comply with the Stockholm Convention and Qatar National Vision 2030 (QNV-2030). The project combines multidisciplinary expertise and aims to provide actionable pollution control guidelines, utilizing Blockchain technology to enhance data security and accessibility.
- **Working Group Member, IEEE Standards Association (SA):** Collaborated on the development of the EDU P3722 Guide for Responsible Artificial Intelligence (AI) in Higher Education. Contributing to a pioneering initiative aimed at establishing ethical and responsible practices for the integration of AI in higher education. Engaged with cross-disciplinary experts to create comprehensive guidelines that support educational institutions in leveraging AI technologies while prioritizing ethical considerations and societal impact.
- I am also involved in another research project with Hail University, KSA, where I serve as the **Primary Investigator**. This research investigates blockchain-based authentication mechanisms to enhance security and privacy in decentralized finance (DeFi). It proposes leveraging blockchain tokenization and NFTs to establish secure digital identities for stakeholders, addressing vulnerabilities in traditional authentication, such as phishing and identity theft. The study combines qualitative (literature review, case studies) and quantitative methods, with a four-phase model covering identity digitization, NFT-based authentication, and deployment on private/public blockchains. Key contributions include improving confidentiality, integrity, and trust in financial transactions while addressing scalability, regulatory, and privacy challenges. The expected outcome is a robust authentication framework that strengthens security and promotes wider adoption of DeFi platforms.
- Furthermore, my co-supervised PhD student in Malaysia is developing a secure and resilient Post-Quantum Resistant Cryptographic (PQRC) system to protect Critical National Infrastructure (CNI)

applications there. The proposed solution explores integrating quantum-resistant cryptographic algorithms designed to withstand quantum computational attacks into a blockchain-based, decentralized, private PKI to secure communications between assets in the CNI. Additionally, the research aims to develop a secure, private, permissioned blockchain-based digital Certificate Authority (CA), reinforced with Post-Quantum Cryptographic (PQC) protocols, to bolster the resilience against advanced cyber threats. The project's expected outcomes are significant and far-reaching, aiming to close current security gaps by contributing to a comprehensive understanding of cutting-edge quantum-resistant cryptography and blockchain technology, addressing existing knowledge gaps, creating a robust private PKI system, and enhancing overall security posture. These efforts will have broader implications for cybersecurity, paving the way for developing and deploying more secure and resilient systems—providing a new generation of security solutions capable of withstanding the most advanced attacks. The anticipated impact of this project is multifaceted. Primarily, it will improve the overall security and resilience of CNI, protecting vital systems and services from cyber threats and potential disruptions, and ensuring the confidentiality, integrity, and availability of critical data and communications.

- In another research project, my PhD student completed his PhD in 2024 on "Blockchain Architecture for NFT-Based Secure Asset Authentication in Cyber-Physical Systems. " In this work, NFTs, non-fungible tokens, were used to authenticate and digitize the assets of IoT devices within CPS. The ERC-721 NFT standard was modified, with new functionalities added. The main goal of the cyber-physical environment is to unify real-time response applications. However, the cyber-physical landscape presents security threats that can compromise data belonging to individuals, systems, and processes. Various centralized and decentralized architectures support smart cities, but concerns remain about traceability, scalability, security, platform support, and resource management. Central to these Systems are IoT-enabled smart devices forming an interconnected network that depends on each other for seamless operation.
- This thesis introduces the Decentralized Smart City of Things (DSCoT), a blockchain-based architecture using both private and public blockchains. DSCoT employs fog computing to manage users and smart devices linked to fog nodes in a cyber-physical environment by extending blockchain tokenization, specifically NFTs. NFTs' unique, non-interchangeable data units on a digital ledger represent distinct assets; however, they cannot effectively digitize IoT assets in smart city cyber-physical systems. The DSCoT architecture extends NFTs to include attributes for Owners, Users, Fogs, and IoT-enabled devices, filling gaps in the ERC 721 protocol. The new NFT mechanism within DSCoT facilitates device-to-device mapping and user authentication for IoT functions. This mechanism aims to provide a secure cyber-physical environment with strong confidentiality, integrity, availability (CIA), and authorization. By introducing new attributes and functions to identify and authenticate Owners, Users, Fogs, and IoT devices, the proposed solution highlights its effectiveness. Performance evaluations concerning cost and time complexity show promising results, validated through experimentation. This study involves enhancing and refining the NFT protocol for cyber-physical systems, then implementing it on private and public ledgers for comprehensive evaluation and comparison. The comparison results demonstrated highly promising improvements in efficiency, cost, and time complexity- up to 96.96.69%- compared to other NFT-based solutions.
- In 2022-2023, I participated in a research grant to develop a secure and resilient Post-Quantum Resistant Cryptographic (PQRC) system to protect Qatar's Critical National Infrastructure (CNI). The research focuses on integrating quantum-resistant cryptographic algorithms, designed to withstand quantum computational attacks, into a blockchain-based, decentralized, private PKI to secure communications between CNI assets. Additionally, we plan to create a secure, private permissioned blockchain-based digital Certificate Authority (CA), strengthened with Post-Quantum Cryptographic (PQC) protocols, to improve the resilience of CNI against advanced cyber threats. The project aims to produce significant and broad-reaching results by addressing current security gaps, advancing understanding of quantum-resistant cryptography and blockchain technology, filling knowledge gaps, developing a robust private PKI system, and enhancing overall cybersecurity.

- Previously, as a research fellow in 2019-2020, I participated in a Blockchain-enabled healthcare project funded by the Ministry of Health, UAE, at Khalifa University, UAE. I helped develop and implement Blockchain-enabled technologies, including Ethereum Smart Contracts, Hyperledger Fabric, Hyperledger Besu, and Quorum platforms for various healthcare-related applications and architectures. Currently, I am working on various decentralized and distributed applications to offer solutions for supply chain industries, including drug and vaccine traceability, healthcare architectures, hospital management systems, patient health record management, and digital identity systems using distributed ledger technologies. Major issues and challenges that Blockchain-enabled distributed ledger solutions address include security, privacy, provenance, and scalability.
- Before this, I was an active research member of the Cybersecurity and Information Security Research Group at Effat University in Jeddah, KSA, from 2015 to 2019. Earlier, during my Postdoctoral fellowship at the International Islamic University Malaysia from 2013 to 2015, I led a vibrant and cohesive research group (Co-PI with Prof. Dr. Asadullah Shah) focused on "Designing and implementing secure & energy-efficient algorithms for cloud-enabled infrastructures.
- Additionally, I participated in and chaired multiple international conference sessions and served as a member of the technical program committee at the 7th International Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS 2025, 2024, 2023, 2022, and 2021), and the 4th Blockchain and Internet of Things Conference (BIOTC 2022). I was the "Scientific Committee Chair" at the 16th Learning & Technology Conference 2019 (Elsevier Indexed) held at Effat University in Jeddah in 2019. I also served as a Program Committee Member for the 5th International Conference on Higher Education Advances (HEAd'19 – HEAd'2026). Additionally, I was on the Scientific Advisory Board for the 4th Renewable Energy Sources - Research and Business conference (RESRB), scheduled for July 8-9, 2019, in Wrocław, Poland. I was the "Track Chair ICT for sustainability" at the International Conference on Information and Communication Technology for the Muslim World 2013 (ICT4M 2013) in Rabat, Morocco, from March 25-27, 2013. I also served as the "Track Chair e-Service/ Ad hoc and Sensor Networks/Internet and web Applications" at the World Congress on Computing and Information Technology (WCIT) 2013 at Asia Pacific University of Technology and Innovation (APU) in Kuala Lumpur, Malaysia, from March 4-6, 2013. Lastly, I was the "Track Chair Wireless Communication and Data Mining" at WCIT 2014 at Asia Pacific University of Technology and Innovation (PAU) in Kuala Lumpur, Malaysia, from March 18-20, 2014.
- My research rigor, hard work, strong leadership, and excellence in research and teaching have given me a solid global network of academic and industry partners for ongoing collaborations. In February 2015, I secured an Internal Research Grant (30000 MR) from the University Malaysia Pahang (UMP) titled "Evaluating Power-Efficient Algorithms for Energy Efficiency and Carbon Emissions in Cloud Data Centers" to further develop my postdoctoral research. The results of this grant received high recognition and were published in Q1 journals. In March 2016, at Effat University, I received another Internal Research Grant (20000 SR) for my research titled "Energy Efficient Multipath Routing Protocol for Mobile Ad-hoc Networks using Fitness Function," which I completed on time. Additionally, I participated in several research projects related to IoT applications and their security infrastructures at Effat University. I also received another Research Grant titled "Robust Security Infrastructure against IoT device-based DDoS Attack in Cloud Environments." I completed it successfully, meeting all the objectives.
- My services to the academic and research community have lasted for nearly 14 years. I have served as a reviewer, guest editor, and editorial board member for various journals and conferences. I regularly review about 15 to 20 research papers each year for reputable journals and conferences, including MISQ, JAIS, IEEE Cloud Computing, IEEE Parallel & Distributed Systems, IEEE Systems Journal, RSER, IJGW, IJGE, AJAIP, Telecommunication Systems, Ecological Economics, EJIS, Computing Journal, IEEE Access Journal, and many others. Additionally, I have chaired sessions, served as a panel member, and acted as an external examiner for several Master's and PhD defenses at Asia Pacific University KL, Malaysia, University Malaysia Pahang, Binary University KL, Malaysia, Effat University, KSA, and Universiti Brunei Malaysia.

- Additionally, I have been extensively involved in designing and developing curricula for several universities I have worked with and have made a significant contribution to the global Computer Information Systems and Cybersecurity curriculum. I currently serve on the College and University committees and the College Academic Promotion Committee at the University of Doha for Science and Technology, Qatar (2023–present). Since my first appointment in 2012 at Asia Pacific University, KL, Malaysia, I have been solely responsible for developing and delivering various modules for different computer science program streams at both the graduate and undergraduate levels. During that time, I, as a faculty member, partnered with EMC, one of the early cloud-providing vendors in KL, Malaysia, to develop and teach newly introduced cloud computing-related modules. We used their tools to perform labs related to implementing cloud-based solutions in different enterprises.
- In recognition of my research and publication skills, I was appointed Research Coordinator for my faculty in June 2014 to oversee and lead departmental research activities, publish quality papers, and secure local and international collaborations and grants. Later, at Effat University in Jeddah, KSA, I was explicitly assigned to prepare and implement **ABET** and **NCAAA accreditation**-related policies and requirements for our department and university. In 2017, ABET finally accredited our Information Systems department, making us the first private university to earn this accreditation for a specific degree program. During the same period, I was tasked with teaching cybersecurity-related courses. I also drafted and prepared modules and outlines for various cybersecurity courses. Additionally, I led the "Cloud and IoT Security Systems Research Group" in our department, coordinating and developing research activities involving faculty, local and international partners, and students to build cloud-based, energy-efficient solutions grounded in high-performance algorithms, techniques, methodologies, and frameworks for cloud data centers.
- My academic performance is demonstrated by the acceptance of the teaching methodologies I use and adapt in various universities throughout my professional career. These methodologies include outlining case study-based teaching approaches and Program Learning Outcome (PLO) based approaches at APU KL Malaysia, UMP Malaysia, Effat University Jeddah, and UDST Qatar. Finally, my teaching prudence is shown by the introduction and development of cybersecurity certification programs to teach and prepare students, industry professionals, and others from various fields for certifications like CompTIA A+, CISSP, and CISA, which attracted cross-faculty enrollments and many professionals from outside.

EDUCATION & QUALIFICATION

- **Doctor of Philosophy (Information Systems)**
Department of Information Systems
Faculty of Computing
Universiti Teknologi Malaysia UTM (Top 200 University)
(January 2010 - September 2013)
- **Master of Science (Computer Science)**
Isra University, Hyderabad, Pakistan (April 2006 – February 2008)
CGPA: 3.85/4.0
- **Bachelor of Computer Science**
Isra University, Hyderabad, Pakistan (April 2001 – September 2005)
CGPA: 3.12/4.0

UNIVERSITY TEACHING EXPERIENCE & APPOINTMENTS

- **Associate Professor (Data and Cybersecurity)**
July 2022 - To date
College of Computing & Information Technology

University of Doha for Science & Technology
Doha Qatar

Courses Taught: Introduction to Data and Cybersecurity, Computer Networks, Secure Software Development, Cybersecurity Pol. Strat. & Proc., Security Engineering Principles, Digital Forensics, Privacy in Digital Era, Applied Cryptography.

- **Honorary Professor (AI-Cybersecurity)**
April 2024 - To date
Faculty of Information Technology, University of Lahore, Pakistan
- **Honorary Adjunct Professor at NSIT-IFSCS, India**
September 2025 – To date
Institute of Forensic Sciences & Cyber Security, Affiliated to National Forensic Sciences University, Ministry of Home Affairs, Govt. of India)
- **Honorary Research Fellow (Cybersecurity)**
May 2025 – Continue
INTI International University Kuala Lumpur, Malaysia
- **Adjunct Academic/Research Fellow (Cybersecurity and Blockchain)**
June 2022 – May 2024
National Advanced IP6 Center (Nav6)
Universiti Sains Malaysia USM Penang, Malaysia
Research Areas: Blockchain, Metaverse and Smart Cities
- **Adjunct Associate Professor (Cybersecurity)**
September 2022 – January 2023
Department of Computer Science
German University of Technology, GuTech Oman
Muscat Oman
Course Taught: Applied Cryptography
- **Assistant Professor (Cybersecurity and Network Security)
(Postgraduate Program and Research Coordinator)**
January 2021 – June 2022
School of Digital Science, Universiti Brunei Darussalam
Negara Brunei Darussalam
Courses Taught: Introduction to Cybersecurity, Computer Networks, Digital Forensics, Blockchain Technologies, Penetration Testing, and Information Security.
- **Associate Professor (Computer Science)
Head of Teaching Department (HoD)**
June 2019 – June 2020
Faculty of Science and Technology
Ilma University Karachi, Pakistan
Courses Taught: Data & Network Security, Computer Organization & Assembly Language, Advanced Computer Networks
- **Associate Professor (Computer Science)**
June 2020 – December 2020
Department of Computing, Faculty of Computing and IT
Indus University Karachi, Pakistan
Courses Taught: Network Security, Advanced Computer Networks, Wireless Networks.
- **Postdoc Research Fellow**
October 2019 – March 2020
Department of Electrical and Computer Engineering
Khalifa University UAE
Project: IoT-Driven Blockchain Solution for the Management of the Healthcare Supply Chain and Medical Data
- **Assistant Professor (Information Systems)**
October 2015 – June 2019
Department of Information Systems, Faculty of Engineering
Effat University, Jeddah, Saudi Arabia

Courses Taught: IS Fundamentals, IT Infrastructure, Data & Network Security, Cybercrime & Digital Forensics, Risk Management, Governance and Control, Cloud Infrastructures, Information System Security, Business Intelligence and Business Process Modelling.

- **Senior Lecturer (Assistant Professor)**

October 2014 – July 2015
Faculty of Computer Systems and Software Engineering
University Malaysia Pahang

Courses Taught: Network Security, Industrial Networks, Problem Solving in ICT, Web Technologies

- **Senior Lecturer (Assistant Professor)**

May 2012 – October 2014
Faculty of Computing and Technology
Asia Pacific University of Technology and Innovation, Kuala Lumpur, Malaysia

Courses Taught: Cloud Infrastructure & Services, Research Methods for Computing and Technology, Computer Networks, Network & IS Security, Wireless and Mobile Computing

- **Postdoctoral Research Fellow**

March 2013 – April 2015
Kulliah of Information Communication Technology
International Islamic University Malaysia
Project: Cloud-enabled Energy-Efficient Data Centers

- **Research Coordinator (Green IT & Cloud Computing)**

June 2014 – October 2014
Asia Pacific University of Technology and Innovation, Kuala Lumpur, Malaysia
Project: Green Cloud Data Centers & Network Security

- **Visiting Supervisor/External Examiner (Postgraduate)**

December 2014 – Ongoing
Binary School of Computing & Entrepreneurship
Binary University of Management & Entrepreneurship, Kuala Lumpur, Malaysia
Responsibilities: External Postgraduate Examiner and Supervisor (Postgraduate Programs)

- **Lecturer**

April 2007 – December 2009
Faculty of Computer Science & IT
Isra University, Hyderabad, Pakistan
Courses Taught: Computer Networks, Statistics, Introduction to Computers

- **Lecturer & Coordinator (Evening)**

August 2005 – December 2009
New Trends Institute of IT, Hyderabad, Pakistan
Courses Taught: Computer Networks, Programming, Network Security

HONORS, AWARDS & RECOGNITIONS

- **Best Researcher Award** from the University of Doha for Science and Technology (UDST), Qatar, 2024.
- **Working Group Member, IEEE P3722 Standard Development (2025-2029):** Guide for Responsible AI use in Higher Education
- **Certified Expert:** Evaluate grant proposals and industry projects for the European Union (EU) Funding & Tenders Portal (Expert ID: EX2026D1377377).
- **Wiley Top Cited Article** in 2025, "Exploring the Convergence of Metaverse, Blockchain, and AI: A Comprehensive Survey of Enabling Technologies, Applications, Challenges, and Future Directions".
- **Certificate of Appreciation from IEEE**, volunteered as a **Proctor** to guide and oversee competing teams for the **IEEE Xtreme19.0 programming Competition** that hosted +19,000 participants on 25th October 2025.

- **Certificate of Appreciation from IEEE**, volunteered as a **Proctor** to guide and oversee competing teams for the **IEEE Xtreme18.0 programming Competition** that hosted +19,000 participants on 26th October 2024.
- **Outstanding Associate Editor Award** from IEEE Access in 2023.
- **Approved PhD Supervisor** at Higher Education Commission (HEC), Pakistan (2021-Present).
- **Certificate in Teaching Skills Masterclass (2019)** from Effat University, offered by **Advance Higher Education (HE), United Kingdom**.
- **Expert Peer-Review Research Proposals** at "The National Center of Scientific and Technical Evaluation (NCSTE) in Kazakhstan from Kazakhstan's Scientific Community 2020-Present.
- **Best Researcher Award (2019)** from the Dean of the College of Engineering, Effat University, Jeddah, Saudi Arabia.
- **Scientific Committee Award in 2019** for the **16th Learning & Technology Conference in March 2019**, titled "**Machine Learning: Intelligence is Power**," held at Effat University, Jeddah, Saudi Arabia.
- **Certificate of Participation at the 16th Learning & Technology Conference, March 2019, for "Machine Learning: Intelligence is Power"** at Effat University, Jeddah, Saudi Arabia.
- **Certificate of Appreciation (2019) for Teaching & Research** from the Dean of the College of Engineering at Effat University in Jeddah, Saudi Arabia.
- **Certificate of Appreciation (2018) (Teaching & Research)** from the Dean of the College of Engineering, Effat University, Jeddah, Saudi Arabia
- **Certificate of Appreciation (2017) (Teaching & Research)** from the Dean of the College of Engineering at Effat University, Jeddah, Saudi Arabia.
- **Certificate of Appreciation (2017) for Teaching, Research, and Community Service** from the President of Effat University, Jeddah, Saudi Arabia
- **Best Community Service Award (2017)** from Dean of the College of Engineering at Effat University, Jeddah, Saudi Arabia.
- **Certificate of Appreciation (2016) for Teaching & Research** from the Dean of the College of Engineering at Effat University, Jeddah, Saudi Arabia.
- **Certificate of Appreciation (2014-2015) for Teaching & Research** from Asia Pacific University KL, Malaysia.
- **Certificate of Appreciation (2017) as a Mentor for "NASA Space Apps Challenge"** on April 27-29, 2024, at Effat University, Jeddah, Saudi Arabia.
- **Certificate of Participation in the 13th Learning & Technology Conference, February 2016, "3,4,5 Dimensions & Beyond: Immerse into the Learning Environment!"** at Effat University, Jeddah, Saudi Arabia.
- **Certificate of Participation at the 14th Learning & Technology Conference, February 2017, "The Makers Space from Imagining to Making"** at Effat University, Jeddah, Saudi Arabia.
- **Certificate of Participation at the 15th Learning & Technology Conference, February 2018, for "IoT Embedding Intelligence"** at Effat University, Jeddah, Saudi Arabia.
- **Certificate of Attendance** for the **Advising & Active Learning Workshop (2017)** at Effat University, Jeddah, Saudi Arabia.

- **Certificate of Attendance (2017)** for participating in **The Center of Excellence in Teaching & Learning Workshop** at Effat University, Jeddah, Saudi Arabia.
- **UTM ZAMALAH Ph.D. Fellowship** (2010-2011), awarded by Universiti Teknologi Malaysia UTM.
- **International Doctoral Fellowship (IDF)**, awarded by Universiti Teknologi Malaysia, 2011-2013.
- **Partial Support Program Scholarship** (Last year of Ph.D. - 2012) awarded by the Higher Education Commission (HEC), Pakistan.
- **Best Paper Award**, International Journal of Current Research and Reviews (IJCRR), 2011.
- **Best Paper Nomination (2011) at the** International Multi-Conference of Engineers and Computer Scientists (IMECS 2011).
- **Certificate of Appreciation (2013)** awarded by the International Association for Information, Culture, and Industry Technology Malaysia.
- **Session Chair Certificate** for The World Congress on Computing and IT (WCIT) held at Asia Pacific University in Kuala Lumpur, Malaysia, from March 4-6, 2013.
- **Certificate of Session Chair** for the 2nd World Congress on Computing and IT (WCIT) held at Asia Pacific University in KL, Malaysia, from March 18-20, 2014.
- **Session Chair** at the Postgraduate Seminar at Universiti Malaysia Pahang (UMP) on March 23, 2015.

INTERNATIONAL & LOCAL MEMBERSHIPS

- **Senior Member, IEEE** (Membership Number: 91193698) (2021–present)
- **Member of IEEE Consumer Technology Society (CTSoc)** (2025-2026).
- **Member, IEEE Consumer Technology Society's Technical Committee on "Security and Privacy in Consumer Technology (SPC)"**.
- **Member, IEEE Consumer Technology Society's Technical Committee on "Application-Specific CE for Smart Cities (SMC)"**.
- **Member of the University Curriculum and Program Development Committee (UCPDS) at the University of Doha for Science and Technology, Qatar** (2023-2026).
- **Member of the College of Computing and IT Curriculum and Program Development Committee (CCPDS), University of Doha for Science and Technology, Qatar** (2023-2026).
- **Member, Gulf Higher Education (GHE)** (2026-present).
- **Member of IEEE Computer Society** (2021-present)
- IEEE Computer Society Special Technical Community on **Blockchain** (2021–present)
- IEEE Computer Society Special Technical Community on **Cybersecurity** (2021–present)
- IEEE Computer Society Special Technical Community on **Internet of Everything** (2021–present)
- IEEE Computer Society Special Technical Community on **Services Computing** (2021–present)
- IEEE Computer Society Technical Committee on **Security and Privacy** (2021–present)
- IEEE Computer Society Technical Committee on **Computer Communications** (2021–present)
- IEEE Computer Society Technical Community of **Business Informatics and Systems** (2021-present)
- **Member of the Association for Computing Machinery (ACM)** (Membership Number: 1542867) (2020-2022)

- **Member of the Association of Information Systems (AIS) from 2015 to 2022.**
- **Member of the Association of Blockchain Brunei (ABB)** (2021–present)
- Collaborator at FIRST Infrastructure, **Malware Analysis SIG** (2022–present)
- **Senior Member**, International Association of Educators and Researchers (IAER) (Membership ID: 181121)
- **Member of the ES-SIG Association of Information Systems (AIS)** (2015–2022)
- **Senior Member**, International Engineering & Technology Institute (IETI) (Membership ID: 2015082012)
- **Member**, American Association of Science and Technology (AASCIT) (Membership ID: 1001179)
- **Member**, Association of Energy Engineers (AEE) (Membership ID: 78952) (2014–2021)
- **Senior Member**, International Association of Engineering Technology (IAET) (Membership ID: 100823) (2015–2022)
- **Professional Member**, International Association of Engineers (IAENG) (Membership ID: 106968) (2012–2022)
- **Member**, International Association of Computer Science and Information Technology (IACSIT) (Membership ID: 80340689) (2013–2022)
- **Member**, Society of Digital Information & Wireless Communication (SDIWC) (Membership ID: 1304406977) (2013–2022)
- **Student Member**, Institute for Computer Sciences, Social Informatics & Telecommunication Engineering (ICST) (2012–2022)
- **Member**, Institute of Data Science and Artificial Intelligence (2016–present)
- **Student Member** of the Association of Environmental and Resource Economists (AERE) (Membership ID: 3852930) (2013–2022)

INTERNATIONAL SEMINARS, WORKSHOPS, TRAININGS & CERTIFICATES

- Attended QATAR'S CYBER FRONTLINE: DEFEND, ADAPT, LEAD, CYSEC Qatar summit, 8th to 9th September 2026.
- Attended the online **MSAB Mobile Forensics Digital Summit**, strengthening investigative capabilities & contributing to the global digital forensics community, March 10–12, 2026.
- Participated in the **Agentic AI Conference (Future of Data & AI)**, a virtual event from April 6–10, 2026, exploring governance, security, enterprise adoption, and multi-agent systems with industry thought leaders.
- Completed the online course "**Digital forensics**" from **The Open University (OpenLearn)**, passing all mandatory tests and learning to correctly gather, analyze, and present digital evidence to business and legal audiences (Issued: 6 January 2026).
- Completed all program requirements and evaluation criteria administered by **Hack & Fix**, earning the **Certified Phishing Prevention Specialist (CPPS)** certification on December 22, 2025 (Certificate ID: 2107-4239-5532-5365).
- Participated in the **PFIC Winter Session 2026**, a high-impact virtual summit dedicated to advancing expertise in digital forensics, incident response, and OSINT (February 19, 2026).

- Attended the **GCC IS Symposium** held on 20th Nov 2025 at Carnegie Mellon University in Qatar.
- Participated in a two-day **Cyber Security Innovation, Research, and Development (CSIRD)** workshop (CSIRD Program Strategy Development, CSIRD Coordination & Partnerships) on 28-29th April 2025 at (QCRI), Hamad Bin Khalifa University Research Complex, Education City, Qatar.
- Attended the **IEEE STEM and IEEE TRY Engineering Summit** on October 26-27, 2025.
- Attended **the IEEE STEM and IEEE TRY Engineering Summit** on October 26-27, 2024.
- Attended the IEEE Authorship and Open Access Symposium: **"Tips and Best Practices to Get Published"** by IEEE Editors on October 24th, 2024.
- Certificate of Appreciation for Serving as Keynote Speaker at the 1st International Conference on Healthcare Innovation and Smart Systems (ICHISS-2024), organized by the Department of Computer Science & Engineering, Shri Shankaracharya Institute of Professional Management and Technology, Raipur, Chhattisgarh, India, from December 29 to 30, 2024.
- Certificate of Recognition for presenting the paper "A signature and NLP-based Network Traffic Detection Model for SQL Injections for Enhanced Web Security at the 17th IEEE/ACM (UCC 2024) and the 11th IEEE/ACM (BDCAT 2024), organized by the University of Sharjah from December 16-19, 2024."
- Attended The Qatar-France Startup Meetup: **"Bridging Innovation: AI, Quantum and Digital Collaboration between France and Qatar,"** at Le Meridian Hotel in Doha, Qatar, on May 20th, 2024.
- Delivered a keynote talk on **"Web3.0 Unleashed: NFTs for the Authentication of IoT Assets in Cyber-Physical-Systems"** at "The International Conference on Advanced Innovative Research in Engineering and Technology" (ICAIRET 2K24) from March 13 to 15, 2024.
- Delivered a talk on **"Beyond Ownership: NFTs and Blockchain's Impact on the Evolving Metaverse"** at the College of Computing and Information Technology (CCIT) Talk Series at the University of Doha for Science and Technology in Qatar on February 26, 2024.
- Attended **Professional Development Day "Learning Landscape: MOOCs and Micro-Credentials"** at the University of Doha for Science & Technology, March 6-7, 2024.
- Delivered a guest lecture on **"Web3.0 Unleashed: NFTs and Blockchain Leading the Metaverse Evolution"** at the Faculty Training Programme on "Demystifying Artificial Intelligence and its Applications by Blockchain and Cyber Security", at Vellore Institute of Technology (VIT), India, December 2023.
- Attended **Professional Development Day "Working with Industry: Experiential Learning in Practice"** at the University of Doha for Science & Technology, March 8-9, 2023.
- Attended the **"ABET Accreditation Workshop"** at the College of Computing and IT, University of Doha For Science & Technology, on March 9th, 2023.
- Attended **Professional Development Day "Working with Industry: Experiential Learning in Practice"** at the University of Doha For Science & Technology on September 22, 2022.
- Delivered a **keynote speech** on **"Blockchain for Cybersecurity Related Challenges & Issues in Healthcare"** at the 6th Global Conference on Computing & Media Technologies at Asia Pacific University, KL, Malaysia, on June 15, 2022.
- Delivered a **webinar** on **"Cybersecurity Issues and Challenges in Military Cyberspace"** at KD-5103 Asia Strategic Environment Studies module in the Graduate Diploma of Defense and Security Studies (GDSS), Brunei Darussalam, on June 13, 2022.
- Participated in **the ASEAN Cybersecurity Skilling Programme: Focus Group Discussion (FGD)** as a Cybersecurity Professional (Academic and Consultant) on April 6, 2022.

- **Interactive Activities for Online Lecturers**, an online workshop at Universiti Brunei Darussalam, organized by the Teaching and Learning Center (TLC) on April 20th, 2022.
- Participated in the **"IEEE Blockchain Symposium"** on March 9, 2022, at the University of Dubai, UAE.
- Attended the online **"IEEE Quantum Computing Healthcare Summit"** on February 2, 2022.
- Attended the online **"IEEE Quantum Internet Summit"** on February 23, 2022.
- **Guest speaker** at the **closed International Cybersecurity Colloquium** at Universiti Kebangsaan Malaysia, August 19, 2021.
- Training Workshop on the latest Blockchain Technologies (**Ethereum Smart Contracts, Hyperledger Fabric, Hyperledger Besu, Quorum**), **Khalifa University, UAE**, Nov 24-26, 2019.
- **Blended Learning During COVID-19 Outbreaks at Kagawa University, Japan**, and an Online Workshop at Universiti Brunei Darussalam, organized by the Teaching and Learning Center (TLC) on March 31, 2021.
- **How to Design Blended Learning Courses & a Lesson Plan?** An online workshop at Universiti Brunei Darussalam, organized by the Teaching and Learning Center (TLC) on June 23, 2021.
- **Online Learning Pedagogies**: an online workshop at Universiti Brunei Darussalam organized by the Teaching and Learning Center (TLC) on February 24th, 2021.
- **Participated in an online workshop at Universiti Brunei Darussalam on September 8, 2021, focused on using Microsoft PowerPoint for slide presentations and poster creation**, organized by the Teaching and Learning Center (TLC).
- **Online workshop on managing references and academic writing** using Mendeley & Microsoft Word at Universiti Brunei Darussalam, organized by the Teaching and Learning Center (TLC) on November 3, 2021.
- Attended the **Blockchain Industry Dialogue** at DIGI TECH Week 2021 at the International Convention Center Brunei, June 17-20, 2021.
- Participated in the **Cyber Battle: Capture the Flag 2021 Competition** at DIGI TECH Week 2021 at the International Convention Center Brunei, from June 17 to 20, 2021.
- Participated as a **"Working Group Member"** in the **"ASEAN Policy Group Discussion"** at an International Teleconference on **Technology and Policy to support the implementation of the COVID-19 Response and Recovery Plan** on Saturday, August 7th, 2021.
- Participated in the **ITTP COVID-19 Webinar Forum Group Discussion "ASEAN DIGITAL Security & Interoperability Standards"** on Saturday, July 24th, 2021.
- Attended **the ITTP - UNAIR Webinar on "Guidelines for Living Together with COVID-19"** on October 27-28, 2021, at UNIVERSITAS AIRLANGGA, Indonesia.
- **Combining Mendeley and Microsoft Word for Managing References and Academic Writing**: Online Workshop at Universiti Brunei Darussalam, Nov 3, 2021.
- Online Training on **"Hyperledger Fabric Deployment on AWS, IBM, and Oracle platforms,"** Nov 6, 2021.
- Online Training on **"Hyperledger Fabric V2 in Action - Best Practices for HF V2 Migration,"** October 22- November 2021.

- Research Seminar on “**Research Opportunities in (and Using) Cloud Computing from an Industry Perspective,**” January 27, 2010, UTM Malaysia.

EDITOR/ASSOCIATE EDITOR/TECHNICAL PROGRAM COMMITTEE MEMBER

- **Editorial Advisory Board Member**, New Review in Information Networking (IF-2.1) (Feb 2025-present)
- **Associate Editor**, Scientific Reports Journal (IF-4.9) (2024- present)
- **Academic Editor at** PLOS ONE (IF-3.6) (2022–2026)
- **Associate Editor at** IEEE Access (IF-3.6) (2022–present)
- **Associate Editor** for Human-Centric Computing and Information Sciences (HCIS) Journal (Area: Privacy, Security, and Cryptography) (IF-5.700) (2022-present)
- **Associate Editor**, IET Networks (IF-1.4) (2021–2024)
- **Editorial Board Member**, Blockchain in Healthcare Today (Scopus, Q3), (2021–present)
- **Associate Editor** for the Journal of Advances in Information Technology (Scopus) (May 2024–present)
- **Area Editor**, EAI Transactions on Energy Web (Scopus, Q2) (2022-2024)
- **Associate Editor** for Frontiers in Communications and Networks (Specialty Section: Security, Privacy, and Authentication) (2022–present)
- **Associate Editor** for Frontiers in the Internet of Things (Specialty Section: Security, Privacy, and Authentication) (2022-2026)
- **Associate Editor** for Frontiers in High Performance Computing (Big Data and AI) (2024-2026)
- **Lead Guest Editor** SI: AI-Driven Secure Integration of Metaverse, Web3, and Quantum Technologies for Next-Generation Distributed Systems, in Computers Materials and Continue (IF-1.7), 2026
- **Guest Editor SI** “Leveraging Information Systems and Artificial Intelligence for Public Health Advancements,” in Frontiers in Public Health (IF-3.5) 2025-2026.
- **Lead Guest Editor** SI “Revolutionizing Blockchain Technology for Smart and Sustainable Transportation and Logistics, in Sustainability (IF-3.868) 2024.”
- **Lead Guest Editor** SI “Revolutionizing Blockchain Technology for Smart and Sustainable Transportation and Logistics, in Sustainability (IF-3.868) 2024.”
- **Guest Editor** SI “IoT in the Industry Revolution 4.0”, in Electronics (IF-2.671) 2023.
- **Guest Editor** SI “Technological development in Point of Interest Recommendation for Smart and Sustainable Cities,” in Frontiers in Sustainable Cities (IF-2.6), 2023.
- **Guest Editor** SI “Exploring the Frontiers of Security, Privacy, and Trust in the IoT Ecosystem,” in Sensors (IF-3.9), 2023.
- **Guest Editor** SI “Surveys on Security, Privacy, and Authentication in Data, Communications, and Networks”, in Journal of Frontiers in Communications and Networks 2023.
- **Editorial Review Board Member**, International Journal of Decision Support Systems (IJDSS), Journal (Scopus) (2022–present)
- **Editorial Member of the** Asian Council of Science Editors (ACSE) (2018–present)

- **Technical Program Committee (TPC)** Member, IEEE HONET 2026 (2026 IEEE 23rd International Conference on Smart Communities: Improving Quality of Life using AI, Robotics and IoT (HONET)), scheduled at Salamis Bay Hotel, Haspolat, Gazimağusa, Mersin 10, Türkiye.
- **Technical Program Committee (TPC)** Member, 15th IEEE International Conference on Consumer Electronics (ICCE Berlin 2026), 5–7 September 2026, Berlin, Germany.
- **Technical Program Committee (TPC)** Member, 10th IEEE CTSoc International Conference on Gaming, Entertainment, and Media (GEM), 5-7 September 2026.
- **Technical Program Committee (TPC)** Member of the 36th International Telecommunication Networks and Applications Conference (ITNAC 2026), 25 – 27 November 2026, RMIT University, Melbourne, Australia.
- **Technical Program Committee (TPC)** member, 9th IEEEJ International Conference on Image Electronics and Visual Computing (IEVC 2026), Hiroshima University, Hiroshima City, Japan, 16-19 March 2026.
- **Technical Program Committee (TPC)** member, 2026 IEEE International Conference on Consumer Electronics–Taiwan (ICCE-TW 2026), July 1-3, Taiwan.
- **Technical Program Committee (TPC)** Member at the 14th IEEE International Conference on Consumer Electronics, September 6–8, 2025, Berlin, Germany.
- **Technical Program Committee (TPC)** Member of the 35th International Telecommunication Networks and Applications Conference (ITNAC 2025), Canterbury, New Zealand, from November 26th to 28th, 2025.
- **Technical Program Committee (TPC)** Member of IEEE HONET 2025 (2025 IEEE 22nd International Conference on Smart Communities: Improving Quality of Life Using AI, Robotics, and IoT).
- **Scientific Program Committee (SPC)** Member of the 11th International Conference on Higher Education Advances (HEAd'25), June 17–20, 2025.
- **Technical Program Committee (TPC)** Member at the 2025 IEEE International Conference on Emerging Innovations and Digital Technology (ICEIDT), November 24–25, 2025, in the Maldives.
- **Technical Program Committee (TPC)** Member at the 6th International Conference on Data Engineering and Communication Technology (ICDECT-2025), indexed in Scopus, held at Bhubaneswar Engineering College (BEC), India, on December 1-2, 2025.
- **Technical Program Committee (TPC)** Member at the 4th International Conference on Sensing, Measurement, Communication, and Internet of Things Technologies (SMC-IoT 2025), Luoyang, China, November 28-30, 2025.
- **Senior Scientific Committee (SPC)** Member, International Congress on Computer Science (AICCS), April 17-18, 2025, Oxford, United States / Online.
- **Scientific Program Committee (SPC)**, 2024, 10th International Conference on Higher Education Advances (HEAd'24).
- **Technical Program Committee (TPC)** Member, International Conference on Intelligent Systems and Security (ICISS2024) to be held from December 20-22, 2024, at the Indian Institute of Engineering Science and Technology, Shibpur, West Bengal.
- **Technical Program Committee (TPC)** Member, International Organizing Committee of CANWEST: International Research Congress '23, Madrid, Spain, September 11-13, 2023.
- **Technical Program Committee (TPC)** Member, 2023 21st IEEE Student Conference on Research and Development (SCORed), APU KL, Malaysia, December 13-14, 2023.

- **Technical Program Committee (TPC)** Member at the 4th Blockchain and Internet of Things Conference (BIOTC 2022), July 8-10, 2022, Tokyo, Japan.
- **Technical Program Committee (TPC)** Member at the 4th Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS), September 27-30, 2022, Paris, France.
- **Technical Program Committee (TPC)** Member at the 3rd International Conference on Blockchain & Internet of Things (BloT2022), March 26-27, 2022, Sydney, Australia.
- **Technical Program Committee (TPC)** Member at the 2nd International Conference on Information Management, Innovation Management, and Industrial Engineering (ICIII 2022), September 16-18, 2022, Bangkok, Thailand.
- **Technical Program Committee (TPC)** Member at the 5th International Conference on Recent Innovations in Computing (ICRIC-2022), held from May 13 to 14, 2022, at Central University of Jammu, J&K. ICRIC-2022.
- **Scientific Program Committee (TPC)** Member at the 8th International Conference on Higher Education Advances (HEAD'22), June 14-17, 2022, Valencia, Spain.
- **Technical Program Committee (TPC)** Member at the 3rd Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS), September 27-30, 2021, Paris, France.
- **Technical Program Committee (TPC)** Member at the International Conference on Technological Innovation and Social Governance (TISG2021), December 14-16, 2021, Guangzhou, China.

RESEACH PUBLICATIONS & GRANTS

Over the past fifteen years, I have been engaged in rigorous, progressive research in cybersecurity, focusing on Blockchain, the Metaverse, Web3, and other security aspects of IoT and Edge infrastructures. My persistent focus, dedication to thorough research, and diligent results have earned me global recognition. Currently, I am involved in numerous research projects, including securing the chain of custody in Multimedia and IoT Forensics using Blockchain, blockchain applications in the healthcare industry, digital identity systems with blockchain, privacy and secrecy issues for cyber-physical systems, and resilience and dependable computing paradigms.

Summary highlights of my research profile can be demonstrated through:

- **220 – Peer-reviewed papers in leading Journals and Conferences**
- **180 – Scopus-indexed Publications**
- **400 – Cumulative Impact Factor (CIF)**
- **6230 – Scopus Citations**
- **41 – h-index (Scopus)**
- **9130 – Google Scholar Citations**
- **51 – h-index (Google Scholar)**
- **112 – i10-index (Google Scholar)**

International Research Grants

Recognized through Awards and Honoraria

1. **Principal Investigator**, "Next-Generation Protein-Rich Azolla Systems: IoT-Controlled Cultivation, AI Yield Prediction, Post-Harvest Quality Enrichment, and Blockchain-Enabled Sustainability for Qatar Food Security Initiatives," (FSC06-0203-260018), **QRDI** Qatar, Submitted April 2026.

2. **Principal Investigator**, "SHIELD-Q: AI-Powered Pathogen Surveillance and National Disease Database for Smart Greenhouse Management in Qatar", (FSC06-0317-260116), **QRDI** Qatar, Submitted April 2026.
3. **Co-Principal Investigator**, "Securing Decentralized Fintech with Blockchain-Based User Authentication," research proposal No. (**RG-24 149**), January 2025, Hail University, KSA.
4. **Co-Principal Investigator**, "AI-enabled Indoor Fire Detection and Alerting System for Blind and Visually Impaired using Deep Transfer Learning and Smart Glasses, February 2026, Submitted to King Salman Center for Disability Research, KSA.
5. **Principal Investigator**, "Q-POPs: Systematic Monitoring of Qatar's Persistent Organic Pollutants to Comply with Stockholm Convention and QNV-2030's Environmental Stewardship Targets," (CCEC02-0217-250069), **QRDI** Qatar, Submitted May 2025.
6. **Principal Investigator**, "Predicting Students' Performance using Artificial Neural Network-based Intelligent Model", Universiti Brunei Darussalam, **UBD/RSCH/1.18/FICBF(a)/2022/003**, August 2021-May 2022.
7. **Principal Investigator**, "Next-generation Blockchain-Enabled Virtualized Cloud Security Solutions: Review and Open Challenges," Universiti Brunei Darussalam, **UBD/RSCH/1.4.FICBF(a)/2021/010**, April 2021 – December 2021.
8. **Principal Investigator**, "On the Design of Lightweight Authentication, Signcryption, and Privacy Preservation Solutions for the Internet of Things," **FIC Research Grant**, Universiti Brunei Darussalam, May 2021 – May 2022.
9. **Co-Principal Investigator**, "Big Data Analytic Model for Irregular Traffic Detection in High-Speed Network, **UKM-SPKP-CRIM-PK01-BO11**, Universiti Kebangsaan Malaysia UKM, November 2022.
10. **Principal Investigator**, "Hadoop-based Big Data Security Infrastructure against DDoS-based Dyn and MQTT Attacks in Cloud Environments (**UC#7/28.FEB.2018/10.2-44(e)**), Internal Research Grant from Effat University, Jeddah, Saudi Arabia, March 2018 - May 2019.
11. **Principal Investigator**, "Multipath Routing Protocol for Mobile Ad-hoc Networks," (**UC#7/02.MAR.2026/10.2-20a**), Internal Research Grant from Effat University, Jeddah, Saudi Arabia, Feb 2016 - Mar 2017.
12. **Principal Investigator**, "Robust security infrastructure against IoT device-based DDoS attack in cloud incentives," Internal Research Grant from Effat University, Jeddah, Saudi Arabia, Feb 2018 - Mar 2019.

International Research Publication

I have authored more than **220** publications in international peer-reviewed journals and conferences, which attest to my research productivity, efficiency, and consistency. The significant citations in reputable publication outlets within a relatively short period demonstrate the rigor and potential of my research.

International Publications

1. Kamran A.A., Sonia K., **Mueen Uddin**, Fuhid A., Meshari A., and Mohmammd S., "TwinSoft: Risk-Bounded Digital-Twin Control with Semantic Telemetry and SDN/NFV Enforcement for Low-Latency Autonomous Vehicular Communication Networks", Vehicular Communications (IF-6.5), Volume 62, December 2026, 101077.

2. Kamran A.A., Sonia K., **Mueen Uddin**, Meshari H.A., Hamed A., Mohammed A.K., "Coherence-Driven Edge Intelligence for Trustworthy Vehicular Low Altitude Internet of Things Networks", IEEE Internet of Things Journal (IF-8.7), Vol.13, No.16, pp. 37970-37977, August 2026.
3. Kamran A.A, Khursheed A., **Mueen Uddin**, Meshari H., "QED-Net: Quantum Emotional Dynamics Synthesis Network for Sentiment Analysis in Medical IoT", IEEE Transactions on Computational Social Science (IF-4.9), Vol.13, No.3, pp. 4012-4021, June 2026.
4. Kamran A.A, **Mueen Uddin**, Korhan Cengiz, Maha A., Raed A., and Hamed A., "PrivNet—Generative AI-Augmented Quantum Privacy Framework for Vehicular Networks", IEEE Transactions on Intelligent Transportation Systems (IF-9.1), Vol. 27, No.6, pp. 6984-6993, June 2026.
5. **Mueen Uddin**, Sonia K., Fuhid A., Mohammad A., Meshari H.A., Maha M.A., "Hierarchical Privacy-Preserving Security Framework for Real-Time Protection in SAGIN-Enabled Industrial IoT Systems", Neural Networks (IF-7.2), Vol. 201, 2026, 108876.
6. **Mueen Uddin**, Shams Ul Arfeen Laghari, Fuhid Alanazi, Saddam Hussain, Tehseen Mazhar, and MD Arafatur Rahman, "A Critical Analysis of Generative AI: Challenges, Opportunities, and Future Research Directions", Archives of Mathematical Modelling in Engineering (IF-12.9), Vol.33, No.2, pp. 1763-1793, 2026.
7. Wahib J.K., Weifeng S., Meshari H.A., **Mueen Uddin**, Muhammad S.A., Naqash Y., Hybrid Attribute-Based Access Control Framework for Intelligent Computing in Consumer IIoT", IEEE Transactions on Consumer Electronics (IF-9.9), Vol.72, No.1, pp. 1615-1622, 2026.
8. Sonia K., **Mueen Uddin**, Maha A., Khalid M., Muhammad S.A., Raed A., Mahmood ul Hassan, "AMEF: Adaptive Multimodal Engagement Framework for Metaverse with 6G-enabled IoT", Cluster Computing (IF-5.5), Vol.29, No.4 (256), May 2026.
9. Kamran A.A., **Mueen Uddin**, Saif A., Hashem A., Dheyaaldin A., "TUB-IoT: Quantum Trust-Based Big Data UAV Architecture for Security-Centric Internet of Things", IEEE Open Journal of Communication Society (IF-6.2), Vol.7, pp.1235-1248, Feb. 2026.
10. Peer Azmat S., Mohsin I., **Mueen Uddin**, Abdulhalim D., Hassan O.A., "DQ-VeDT: A Decentralised Quorum-based Authentication for Quantum-Safe Vehicular Digital Twins," IEEE Open Journal of Vehicular Technology (IF-6.6), Vol.7, pp. 1814 – 1831, June 2026.
11. Abdulhalim D., Viswa K., **Mueen Uddin**, Usman K., "LLM-Driven Multi-Agent Framework for Autonomous Network Slice Management: Architecture and Proof of Concept Experiments, published in early access in IEEE Open Journal of Communication Society (IF-6.2), Vol.7, pp.7118-7135, July 2026.
12. Anam M., Qasim A., Ali A.M., Meshari H.A., **Mueen Uddin**, Mohammad S., "A Novel Monocular 3D Human Pose Refinement Method using Anthropometric Constraints and Synthetic Localization Errors", The Visual Computer (IF-3.4), Vol.40, No.1, 2026.
13. Chin-Ling Chen, Chenxi H., **Mueen Uddin**, Praveen N., and Minglian Q., "Editorial: Leveraging Information Systems and Artificial Intelligence for Public Health Advancements", Frontiers in Public Health (IF-4.1), 14:1789559, Feb 2026.
14. Usman, K., **Mueen Uddin**, Owais, M., Ong, W.H., "Extending ERC721: Design and implementation of a novel secure NFT framework for IoT asset authentication in cyber-physical systems", Blockchain Research and Applications (IF-6.5), Vol.72, No.2, 100456, May 2026.
15. **Mueen Uddin**, Maha A., Fuhid A., Mohammad A., Raed A., Korhan C., "AMRF—Adaptive Multi-Resolution Fusion for Enhanced Salient Object Detection Using Deep Learning", PeerJ Computer Science (IF-2.5), 12:e4027, DOI 10.7717/peerj-cs.40276, August 2026.

16. Kamran A.A., **Mueen Uddin**, Meshari H.A., Mohammad N.A., Faris A., Mohammad A.K., "ASIF: An Adaptive Swarm Immunity Framework for Anomaly Detection in Smart Home Consumer Electronics", **available in early access** in IEEE Transactions on Consumer Electronics (IF-9.9), September 2026.
17. Momina B., Kamran A.A., **Mueen Uddin**, Chin-Ling Chen, Ling-Chun Liu., Fuhid A., Meshari A.A., and Tahani Alsubait, "EdgeGuard: Cross-LLM Robustness Evaluation of Lightweight Phishing Detectors Against LLM-Generated Email Attacks in Edge-Constrained Environments", **Accepted** in Journal of King Saud University-Computer and Information Sciences (IF-6.4), June 2026.
18. Abdul M., Syed Shayan A.S., Saeed Q., **Mueen Uddin**, Amina S., "PPTM-IoV: A Privacy-Preserving Trust Management Framework for the Internet of Vehicles (IoV)", **Accepted** in Scientific Reports (IF-4.9), July 2026.
19. Saeed Q., **Mueen Uddin**, and Abdul M., "TrustHealth: A Secure and Dynamic Trust-Management Model Using Biometric and Behavioral Data in IoT-based Smart Healthcare Systems", **Accepted** in Informatics for Health and Social Care (IF-1.8), August 2026.
20. Ashraf, I., Kehkashan, T., Rehman, A., **Mueen Uddin**, Akhunzada, A. (2026). A Multi-agent Reinforcement Learning Approach for Seamless and Safe Autonomous Vehicle Platooning. In: Proceedings of the 3rd International Conference on Intelligent Technology, System and Service for Internet of Everything. ITSS-IoE 2025. Lecture Notes in Electrical Engineering, vol. 10154, Springer, Singapore.
21. Abdullah, M., Kehkashan, T., Riaz, R.A., **Mueen Uddin**, Akhunzada, A. (2026). Surface Crack Detection: Advances in Automated Visual Inspection for Infrastructure Safety. In: Proceedings of the 3rd International Conference on Intelligent Technology, System and Service for Internet of Everything. ITSS-IoE 2025. Lecture Notes in Electrical Engineering, vol. 10154, Springer, Singapore.
22. Hadiqa R., Marwa A., Muram K., Eiman I., Zobia A., **Mueen Uddin**, "Blockchain-Based IP Identity Management System for University Intellectual Property Governance", Accepted in International Conference on Emerging Trends in Computing Systems (ICETCS 2026).
23. Saddam H., **Mueen Uddin**, Nayab, "RegimeMoE: Heterogeneous Mixture-of-Experts for Electricity Load Forecasting under Structural Breaks", Accepted in International Conference on AI-IoT Technologies for Renewable and Sustainable Energy (AIoT-RSE 2026).
24. Saddam H., **Mueen Uddin**, Nayab, Hashem A., "Label Leakage in Interest Flooding Attack Benchmarks", submitted in CCTS 2026 (3rd International Conference on Creativity, Technology and Sustainability, 2026).
25. Saddam H., **Mueen Uddin**, Nayab Hashem A., "Evaluation of Computational Complexity of Attribute-based Proxy Signature", submitted in CCTS 2026, 3rd International Conference on Creativity, Technology and Sustainability, 2026.
26. **Mueen Uddin**, Saddam H., Anjum K.B., Wagdi A., Saeed M.Q., Abdulhalim D., and Shams Ul.A.L., "Agentic AI in Cybersecurity: Autonomous Threat Intelligence, Ethical Challenges, and Future Directions", submitted in IEEE Open Journal of Computer Society (IF-6.2), July 2026.
27. **Mueen Uddin**, Hidayatullah K, Saddam H., Maha A., Homam E., Abdulhalim D., "Machine Unlearning as a Security Problem: Threats, Defenses, and the Privacy-Security-Utility Trilemma", **Revision submitted** to Computer Science Reviews (IF-17.5), June 2026.
28. Jamshed M., **Mueen Uddin**, "InterNull: Quantum-Safe Blockchain Accounts Without Post-Quantum Primitives", submitted in IEEE Transactions on Information Forensics & Security (IF-8.7), April 2026.
29. Muhammad Q., Kamran A., Meshari, H.A., **Mueen Uddin**, Muhammad S.A., Muhammad A.K., "EIHCS: Ethical Interference-Driven Human-Centric Scheduling for Secure Social Data Wireless Systems, submitted in IEEE Transactions on Computational Social Systems (IF-4.9), May 2026.

30. Saddam H., **Mueen Uddin**, Anand N., Meshari H.A., Faris A., Mohammad S., "SAND: A Schema-Agnostic and Training-Free Framework for Interest Flooding Attack Detection Across Heterogeneous Named Data Networks", submitted in IEEE Transactions on Networking (IF-5.3), July 2026.
31. Muhammad Q., Kamran A.A., **Mueen Uddin**, Meshari H.A., Maha A., Latifah A., Haya A., "WARDEN-Witness Anchored Relational Detection and Evidence Neutralization for Misbehavior Prevention in Vehicular Networks", Revision submitted to Transactions on Emerging Telecommunication Technologies (IF-2.6), June 2026.
32. Saqib M., Mazhar H.C., Ghulam D., Shaikh A.A., Anand N., and **Mueen Uddin**, "HDLB: A Post-Quantum and Blockchain-Enabled Hierarchical Federated Framework for Secure and Scalable Authentication in Financial Information Systems", submitted in Computer Standards and Interfaces (IF-4.6), March 2026.
33. Kamran A.A., Sonia K., **Mueen Uddin**, Mohammad S., Hamed A., Meshari H.A., Mohammad A.K., "Resource-Aware Quantum-Cognitive Micro Models for Low-Latency Inference in IoT Edge Networks", submitted in IEEE Internet of Things (IF-8.7), June 2026.
34. Kamran A.A., Sonia K., **Mueen Uddin**, Amani I., Tahani A., Mohammad S., Ahmed F., "Biologically Inspired Distributed Learning for Secure Efficient Resource Allocation in Autonomous Vehicles," submitted in Robotics and Autonomous Systems (IF-6.0), Feb 2026.
35. Kamran A.A., **Mueen Uddin**, Mohammad S., Khalid M., Mahmood Ul H., Meshari H.A., Ahmed F., "Energy-Efficient Resource Optimization Framework for Connected Vehicular Engineering Systems with Edge-Cloud Intelligence, submitted in Ain Shams Engineering Journal (6.2), April 2026.
36. Huzaifa B.S., Muhammad M., Kamran A.A., Hamed A., Meshari H.A., and **Mueen Uddin**, "Sentivo: A Real-Time Edge ASR-Bi-GRU-Capsule Framework for Customer Voice Sentiment Analysis and Smart Retail Decision Support, submitted in Measurement (IF-6.1), June 2026.
37. Muhammad Q., Kamran A.A., Meshari H.A., Anand N., Sumendra Y., and **Mueen Uddin**, "Cybersecurity-Driven Communication Framework for Edge-Assisted Satellite Internet of Things Environmental Monitoring", submitted in Connection Science (IF-3.5), August 2026.
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Professional Service

I serve as an Editorial Board Member and Reviewer for various international peer-reviewed journals and conferences, demonstrating academic leadership and professional service at the highest level. I am also frequently invited to be a Technical Program Committee member for many globally recognized conferences. Each year, I review approximately 15 to 20 research papers, depending on my busy schedule, for reputable journals and conferences. I have been a regular reviewer for the following distinguished outlets:

- Blockchain: Research and Applications
- Journal of Supercomputing
- Arabian Journal of Science & Engineering (AJSE)
- Renewable and Sustainable Energy Reviews (RSER)
- Ecological Economics
- Telecommunication Systems
- IEEE Access
- IEEE Cloud Computing
- IEEE Transactions on Parallel and Distributed Systems
- IEEE Systems Journal
- Concurrency and Computation: Practice and Experience
- Journal of Medical Internet Research
- EURASIP Journal on Wireless Communications and Networking
- International Journal of Distributed Sensor Networks
- Information Technology Research Journal (ITRJ)
- Journal of Computer Science
- International Journal of Advanced Intelligent Paradigms IJAIP
- Computing Journal Springer
- Journal of Systemics, Cybernetics and Informatics (JSCI)
- International Journal of Global Warming (IJGW)
- International Journal of Green Economics (IJGE)
- Journal of Testing and Evaluation
- International Journal of Wireless and Mobile and Computing (IJWMC)
- WSEAS Transaction on Communications Journal
- The Energy Journal
- WSEAS Conferences

Academic Service

During my academic career, I have actively participated in designing, updating, and reviewing several courses related to cybersecurity, digital forensics, network and information security, and cloud computing across various university roles. I have regularly served on university, college, and departmental curriculum committees, where I helped assess and revise course content to incorporate the latest technological advances, industry requirements, and accreditation standards such as ABET and other national benchmarks. In addition to teaching and curriculum development, I have supervised numerous final-year projects for both undergraduate and graduate students. Specifically, I have supervised seven Master's students for their theses and forty undergraduate students for their final-year projects. Below is the list of students I have supervised and examined over the past three years.

- Supervising (3) PhD Students
- Supervised (8) Master's Dissertations
- Supervised (40) undergraduate final-year Projects
- External examiner of Master's and Doctoral Theses and Dissertations

PhD SUPERVISION

- Blockchain Architecture for NFT-based Secure Asset Authentication in Cyber-Physical Systems (Co-Supervisor) (2021-2024).
- Quantum-Resistant and Blockchain-Enabled Secure Storage and Transfer of Patient Data in Smart Hospitals (Co-Supervisor) (2023-continue).

- A Decentralized Web3 Architecture for Secure and Scalable IoT Device Authentication in Smart Healthcare Systems (Co-Supervisor, 2023–Present).

MASTERS DISSERTATIONS SUPERVISED

- Detecting Risky Shadow AI Usage in Enterprise Networks Using Network and Behavioral Analysis (2026-2027).
- Agentic AI-Enabled Cybersecurity Framework for Threat Modeling in Cyber-Physical Systems (2025-2026).
- Green ICT Framework for Implementing Sustainable APU Campus at APU KL, Malaysia, 2014.
- Performance Analysis of Mobile Ad-Hoc Network through TCP Variants over DSR, at APU KL, Malaysia, 2014.
- Evaluation of Power Convergence Algorithms using Cloudsim Simulator in Cloud System at APU KL, Malaysia, 2014.
- Comparative Analysis and Evaluation of Vyatta and Nexus Virtual Security Appliance in a Cloud System at APU KL, Malaysia, 2013.
- Cloud computing & Green ICT adoption model for effective green practices in Malaysian Universities at APU KL, Malaysia, 2013.
- Mobile Agent-based Multi-Layer Security Framework for Cloud Data Centers at APU KL, Malaysia, 2013.

EXAMINER DOCTORAL THESIS (PhD & DBA)

- LIGHTWEIGHT AND EFFICIENT NEURAL ARCHITECTURES FOR VISUAL RECOGNITION, at Vellore Institute of Technology, Vellore Campus, India, 2026.
- Protecting Smart Homes: A Study on AI Approaches to Detect Malware and Developing a Two-Factor Authentication System Based on SRAM PUF For Smart Home Users in the UAE, at British University Dubai, 2025.
- Developing a Multivariate Time Series Analysis Using Encoder–Decoder Based Attention Mechanism, VIT Vellore, India (2025).
- Advancing Multivariate Time Series Forecasting Using Hybrid Dimensionality Reduction and Uncertainty-Aware Deep Learning, VIT Vellore, India (2025).
- Efficient Channel Estimation in MIMO Communication Systems using Deep Ensemble Learning at VIT University, Vellore, India, 2025.
- Resource Management in 6G For Vehicle-To-Everything (2025).
- Design of Blockchain-empowered Lightweight Defense Framework for Multi-Vector DDoS Attacks in IoT Networks at USM, Malaysia, 2024.
- Investigations on Deep Learning Techniques in Protein Secondary Structure Prediction at VIT, Vellore, India, 2023.
- An Investigation into the Cyber Security Preparedness of the Zambian Financial Sector, at Binary University KL, Malaysia, 2016.

EXAMINER MASTERS DISSERTATIONS

- Smart Surveillance – Anomaly Detection in Security Camera Feeds, UDST, 2026.

- Visual-Based Phishing Detection Through Website Screenshot Analysis, UDST, 2026.
- Detecting Zero-day Attacks using AI and Machine Learning at Zayed University, UAE, 2025.
- Beyond the Headlines: AI-Driven Detection of Arabic Fake News at Zayed University, UAE, 2025.
- Unveiling the Mastery of Multi-Digit Number Recognition: A Deep Dive into MNIST Dataset for Handwritten Recognition for Digital Forensic Investigations at Zayed University, UAE, 2024.
- A Study on Consumer Attitudes to Lifebuoy 1 Liter Total 10 Bodywash in the Klang Valley at Binary University, KL, Malaysia, 2018.
- Effectiveness of Sales Promotion of Unilever Breeze Liquid Detergent at Binary University, KL, Malaysia, 2018.
- Framework for Streamlining Disaster Recovery Practices for Core Banking Services among Commercial Banks in Sri Lanka at APU, KL, Malaysia, 2014.
- A framework of green technology and power efficiency management in APSS data center for reducing power consumption at APU, KL, Malaysia, 2014.

DEGREE STUDENT'S PROJECTS SUPERVISED

- Detecting and Visualizing Phishing Attacks in the Metaverse: A Threat Intelligence Dashboard (2025-2026).
- Secure Identity Management: Leveraging Blockchain and NFTs for Intellectual Property at UDST (2025-2026).
- A Comparative Analysis of ML Models for Detecting DDoS Intrusions from IoT Botnets in Cloud Environments (2025-2026).
- ChainCare ID: Streamlined Decentralized Identity Management for Scalable Hospital Access at UDST Qatar, 2025). (**Published in Springer Lecture Notes Series**)
- CNN-LSTRM: Hybrid Deep Learning Model for Ransomware Detection in Healthcare at UDST Qatar, 2025).
- NexusPlus: Career Fair 3D-Metaverse with Decentralized Security and AI-based Recommendation System at UDST Qatar, 2025 (**Best FYP Project Award**).
- A Signature and NLP-based Network Traffic Detection Model For SQL Injections for Enhancing Web Security at UDST Qatar, 2025 (**Best FYP Project Award**).
- Patient-Centric Secure and Efficient Electronic Health Record System using Blockchain at UDST Qatar, 2024 (**Best FYP Project Award**).
- "HomeOnUs" web-based system for providing home services in Qatar at UDST Qatar, 2025.
- Secure and Efficient Electronic Health Record Management System using Hyperledger Besu at SDS UBD Brunei, 2021.
- Blockchain-enabled digital forensics investigation framework for IoT infrastructures at SDS UBD Brunei, 2021.
- Secure Electronic Funds Transfer over the Internet using efficient Encryption techniques at SDS UBD Brunei, 2020.
- Green ICT framework for Data Center in Effat University, Jeddah, KSA, 2019.
- Green BPM: An Urban Framework for Assessing & Improving Sustainability in Water Management in KSA at Effat University, Jeddah, KSA, 2019.
- Cloud-based Education as a Service (E-as-a-Service) Framework for Effat University, Jeddah, KSA, 2018.
- Android-based Math Game for kids in school at Effat University, Jeddah, KSA, 2018.

- Online Volunteer Management System for Effat University, Jeddah, KSA, 2017,
- School Management System for Effat University, Jeddah, KSA, 2017.
- Cloud-based Hospital Emergency & Response System at Effat University, Jeddah, KSA, 2017.
- Online Internship Portal System for Effat University Students at Effat University, Jeddah, KSA, 2016.
- "NuCAJ" – Numerical Code Algorithm for Japanese Text Steganography at APU, KL, Malaysia, 2014.
- "C-Mors" Public Cloud Monitoring and Recovery Systems at APU, KL, Malaysia, 2014.
- Train tracking system using data from crowdsourcing at APU, KL, Malaysia, 2014.
- "VBHAS" Voice-Based Home Automation System at APU, KL, Malaysia, 2014.
- "Wi-Fi Real-time Inspector" a Network Management System at APU, KL, Malaysia, 2014.
- Packet Analyzer and Reporting System at APU, KL, Malaysia, 2014.
- Network Protocol Analyzer – Traffic Monitoring through Packet Capturing (PCAP) at APU, KL, Malaysia, 2014.
- Network Traffic Monitoring with Packet Sniffing at APU, KL, Malaysia, 2014.
- Green Computing in Data Center Infrastructures to measure their Performance using PUE Metrics at APU, KL, Malaysia, 2014.
- File Transfer in Client-Server Network using Chain Distributed Transfer Protocol in Wireless Networks at APU, KL, Malaysia, 2014.
- Preventing Misuse of Network Resources in Strengthening Network Management using Wake On LAN Application and Embedded Free Meter at APU, KL, Malaysia, 2014.
- AI-based Network Monitoring Tool-Enhancing Network Security Performance using Intelligence Knowledge Base at APU, KL, Malaysia, 2014.
- Encrypted File Transfer on Android via FTP and PHP Server at APU, KL, Malaysia, 2013.
- Text File Encryption Using XML at APU, KL, Malaysia, 2013.
- An application to hide Digital Watermarks and Metadata on the Noise of JPEG Images for Copyright Purposes at APU, KL, Malaysia, 2013.
- Signature-based Campus Network Analyzer with Packet Filtering at APU, KL, Malaysia, 2013.
- Host-Based Intrusion Detection System for Denial of Service (DoS) Attacks at APU, KL, Malaysia, 2013.
- Enhanced QR code Security for Android mobiles at APU, KL, Malaysia, 2013.
- Detecting Cross-Site Scripting Attacks in Social Networks at APU, KL, Malaysia, 2013.
- Iterative file encryption on Cloud Engineering at APU, KL, Malaysia, 2013.
- Signature-based Campus Network Analyzer with Packet Filtering at APU, KL, Malaysia, 2013.
- An intrusion detection system in a mobile Ad hoc network at APU, KL, Malaysia, 2013.
- Cybercafé Network Monitoring System in Gambian at APU, KL, Malaysia, 2012
- Automatic network pen-testing system at APU, KL, Malaysia, 2012
- Intelligent Factory Ordering and Customization Systems for SME's in Yemen at APU, KL, Malaysia, 2012.
- Data Security System Using Steganography (DSSUS) at APU, KL, Malaysia, 2012.

During my academic career, I have been responsible for the following specific activities:

- **Postgraduate Admission Coordinator** at the School of Digital Science, Universiti Brunei Darussalam (2021-2022).
- **Research Coordinator** at the School of Digital Science, Universiti Brunei Darussalam (2021-2022).
- Providing academic and research training for students and staff.
- Promoting synergy between research and students who work in different research tracks.
- Organizing fortnightly colloquia by arranging venues, speakers, and disseminating appropriate invitations and messages.
- Organizing group reading workshops/seminars for higher degree teaching and research students.
- Representing the university at various levels within the faculty, the University, and in the industry, and uplifting the research quality and quantity of the research tracks.
- Moreover, I have chaired and/or served as a panel member in several oral defenses at both master's and doctoral research levels. I have also served as an external examiner for doctoral dissertations at Binary University, KL, Malaysia.

TEACHING PHILOSOPHY & COURSES TAUGHT

Teaching Philosophy

An effective learning environment requires challenge, respect, and enjoyment. Without challenge, a student may finish a subject feeling vaguely dissatisfied, as if they learned nothing. Without respect, a student might complete a course having resisted genuine learning and wishing to forget the experience quickly. Regardless of the skills the teacher aims to provide, without challenge and respect, those skills will not last beyond the course. The third element, enjoyment, helps facilitate open communication and knowledge absorption, promoting both challenge and respect. I have always strived to incorporate all three principles into my teaching. I encourage students to learn quickly, question what is presented, and seek information beyond the classroom materials. I hold high expectations for their performance and have consistently been pleased with their progress. In return, I constantly look for new ways to teach challenging material and design assignments and projects that are timely, relevant, and interesting. Increasing enjoyment fosters a positive classroom atmosphere conducive to learning and encourages dialogue, helping to address challenges in promoting challenge and respect, especially when some students view them as merely "extra work."

Teaching Experience

I have committed myself to using innovative teaching methods that go beyond the usual duties of a subject coordinator or lecturer. I identified key areas such as methodology, delivery, motivation, and student welfare as ways to improve and elevate your teaching and learning skills. I gained extensive teaching experience through various roles and responsibilities across different subjects by applying these focus areas. I have taught as a lecturer, subject coordinator, and coordinating lecturer in both core and elective undergraduate and postgraduate courses, showing my ability to teach across a wide range of disciplines.

I advocate case-based teaching as an effective method for creating a conducive learning environment. I incorporate the case-based approach in all the subjects I develop and teach. I use this strategy to illustrate and discuss complex theoretical and management aspects of courses like IT Infrastructure, Cybercrime & Warfare, E-Commerce, and Risk Assessment & Management. Additionally, exams and assessments also utilize a case study approach. Case studies allow students to engage with real-world issues and projects that would otherwise be difficult to simulate in teaching these courses. To motivate students to reach their full potential, I introduced the Best Student Awards for my subjects. Since then, I have continued these awards

with strong support from all student cohorts. The top student, based on the highest marks, receives a special plaque and is recognized in the following semester by an industry expert during the lecture.

I have completed the Teaching Skills Masterclass Certificate from Advance Higher Education UK and have learned many new methods of interactive teaching and learning. I am now planning to pursue a fellowship from the same training authority. I have been actively involved in preparing, organizing, and coordinating several courses offered at both undergraduate and postgraduate levels within the Information Systems and Information Sciences cluster. My role typically requires and involves many tasks, including developing course content, preparing lectures and tutorial activities and notes, creating lab materials, assignments, and final projects, recruiting and supervising teaching staff and tutors, selecting readings, setting assessments, moderating exam papers, maintaining the course's online delivery sites (Blackboard and Banner), and liaising with the professional community to arrange guest lecturers. I have taught and prepared the courses listed below, which have been assigned to me in various academic positions over the past approximately 12 years, in accordance with the Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs) requirements at the universities where I have worked. Throughout my academic career, I have taught the following courses at different levels.

- Advanced Information Security (PhD Course)
- Introduction to Theoretical Computer Science (PhD Course)
- Introduction to Data & Cyber Security
- Secure Software Development
- Cybersecurity Policies, Strategies and Procedures
- Applied Cryptography
- Privacy in the Digital Era
- Cybercrime & Warfare
- Digital Forensics & Investigations
- Blockchain Technology & Applications
- Information Security & Policy
- Introduction to Information Security & Management
- Risk Assessment & Management
- Computer Networks
- Discrete structures
- Business Process Modelling
- Data Management & Information Security Audit
- E-Commerce & its Security
- Management Information System
- Network Security
- Advanced Topics in Security Systems
- Information Technology Management
- Ethics & Professionalism in Computing
- Business Intelligence
- Cloud Infrastructure and Services
- IS Strategy & Acquisition
- IT Infrastructure
- Industrial Networks
- Problem-Solving in ICT
- Information Storage and Management
- Research Methods for Computing and Technology
- Wireless and Mobile Technologies
- Mobile Computer Communication Systems
- Fundamentals of Wireless LAN
- Advanced Wireless Networks
- Remote Access Networks
- Design of Corporate Communication Systems

- Operating System
- Advanced Computer Networks
- Computer Organization and Assembly Language

Furthermore, in addition to these courses, I also utilize various tools and software packages for conducting labs and encouraging case study-based learning approaches; some of these tools are listed below.

- Kali Linux
- Wireshark Network Analyzer
- Autopsy Forensic Tool
- NMAP
- FTK Forensic kit
- Volatility
- Rekall
- Encase
- Prodiscover Forensic Tool
- Virtual Box 6.0
- Cisco Packet Tracer
- Configuring Different Servers like ISA, DHCP, IIS
- Network Simulator 2 (NS-2)
- RouterSim Network Visualizer
- Nessus Network Visualizer
- Snort Intrusion Detection System
- PUE and DCE Energy Efficiency Calculators for Data Centers
- Carbon Emission Calculator
- Microsoft Hypervisor V Server
- VMware Capacity Planner for Data Centers
- VMware Guided Consolidation for Data Centers
- Cloudsim Simulator

REFERENCES

Prof. Dr Asadullah Shah
 PhD (Multimedia Systems)
 Kuliyah of ICT
 International Islamic University Malaysia
Asadullah@iiu.edu.my

Prof. Dr Raed Alsaqour
 Professor of Computer Science
 Saudi Electronic University, Riyadh KSA
r.alsaqour@seu.edu.qa

Prof. Chin-Ling Chen
 Department of Computer Science and Information Engineering
 Chaoyang University of Technology, Taichung City, 413310
 Taiwan
clc@mail.cyut.edu.tw

